

Agenda.

- Exceptions
- Hierarchy
- Handling Exceptions (Try-Catch)
- Throw vs Throws.
- Checked vs Unchecked
- String Comparison.

Exceptions.

↳ Something unexpected.

```
int x = sc.nextInt();  
int y = sc.nextInt();  
z = x/y  
scout(2)
```

} Divide By Zero Exception.

Array

```
for (i = 0; i < n+1; i++) {  
    a[i]
```

3

$a[-1]$, $a[n]$ $\Rightarrow$ Array Index Out Of Bound

int x = sc.nextInt()

⇒ InputMismatchException.

⇒

readFromFile (String fileName) {

File f = new File(fileName);

FileReader fr = — ;

fr.read()

3

↓
If filename is invalid
then we can't read the
file content.

⇒ FileNotFoundException

⇒ If we don't handle Exceptions, then our code
will stop abruptly.

Handling Exceptions.

Why? So that our code doesn't crash abruptly.

⇒ try, catch & finally

try { // Code that can potentially throw an exception.

_____ L1

_____ L2

File f = new File(fileName);

FileReader fr = new FileReader(f)

fr.read()

|||

Catch (FileNotFoundException ex) {

|||

Catch (IOException ex) {

|||

Catch (Exception e) {

|||

} The most general catch block should be at the end.

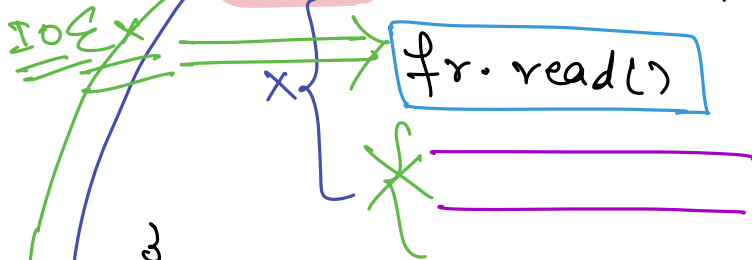
try { // Code that can potentially throw an exception.

✓ ✓ _____ L1

✓ ✓ _____ L2

✓ ✓ file f = new File(fileName);

✓ ✓ FileReader fr = new FileReader(f)



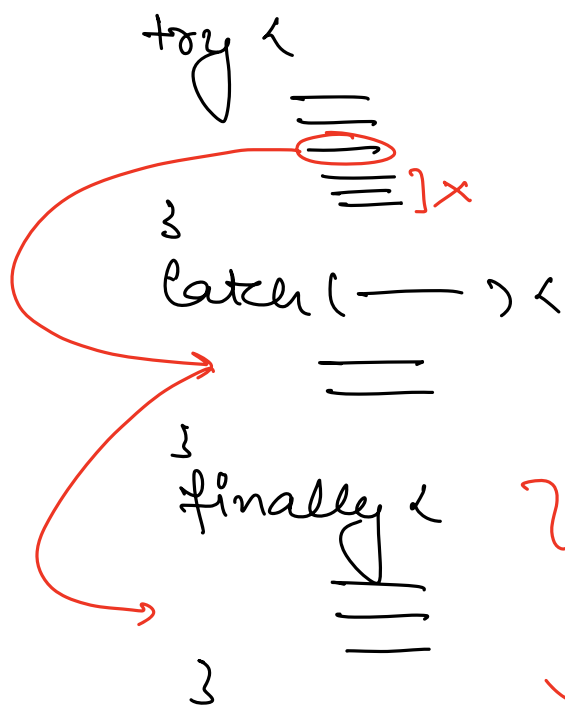
|||
Catch (FileNotFoundException ex) {

✓ {

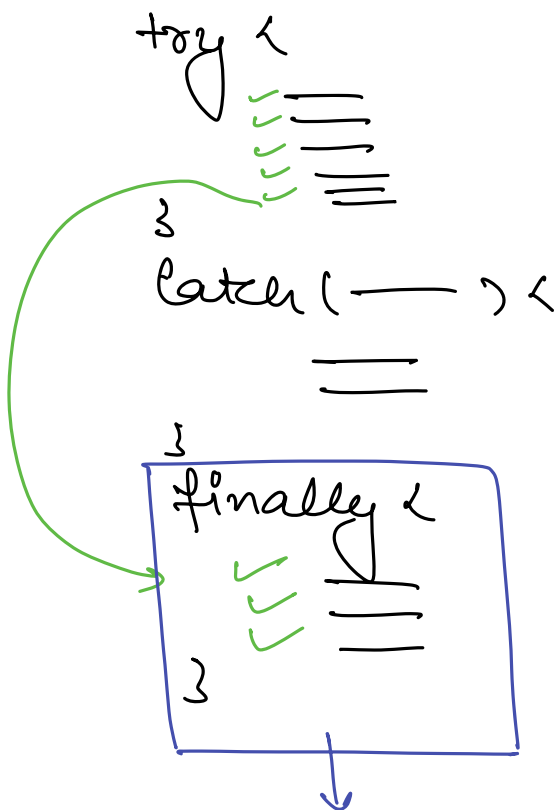
|||
Catch (IOException ex) {

✓ {

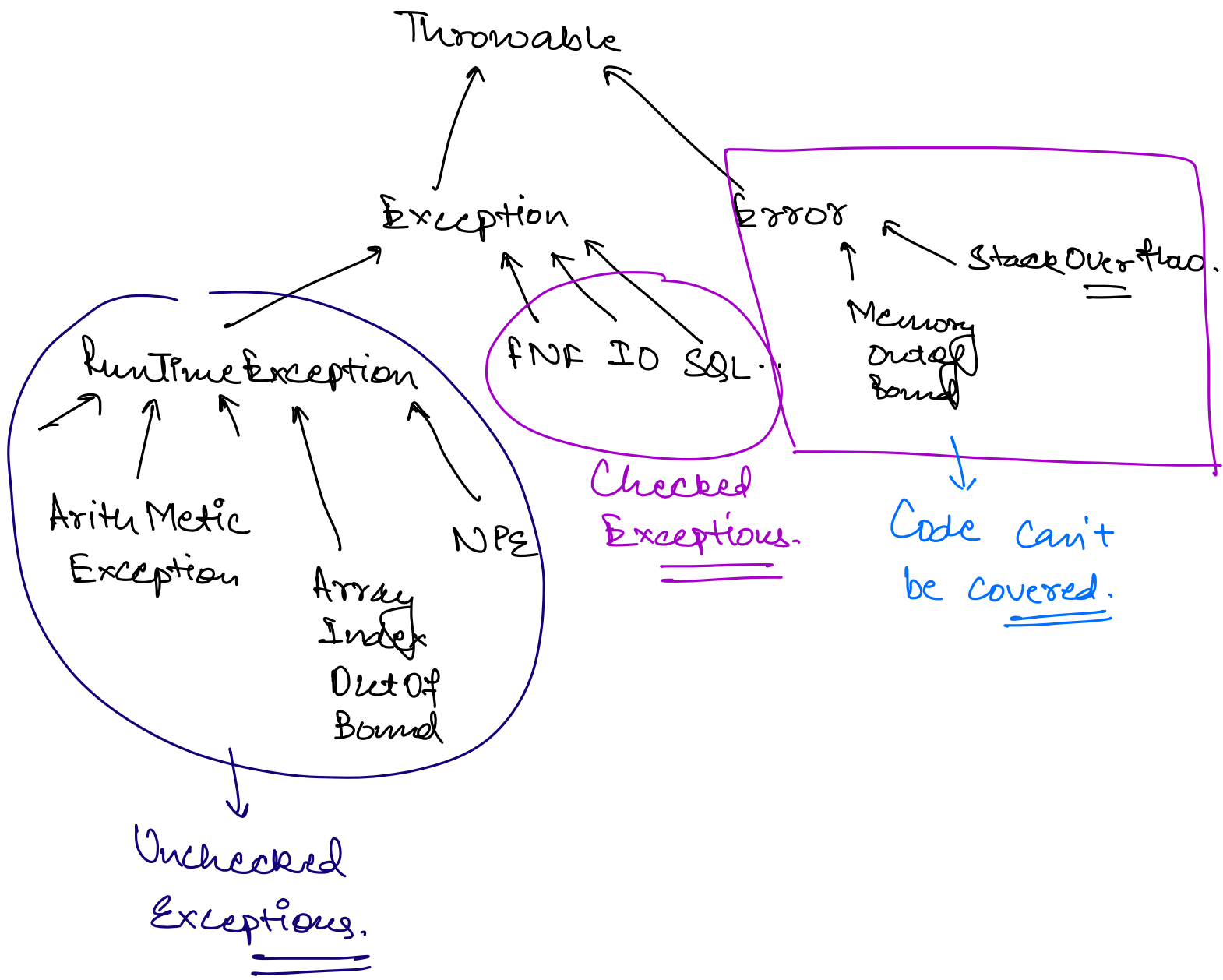
|||
} ✓



} Code that will execute
irrespective of what happens
above.



Close DB connection.



Class UserNotFoundException extends
Exception / RTE {

UserNotFoundException(String msg) {
 super(msg);
}

==

==

Throw

vs

Throws

Method Signature
to pass exception to the
caller funⁿ.

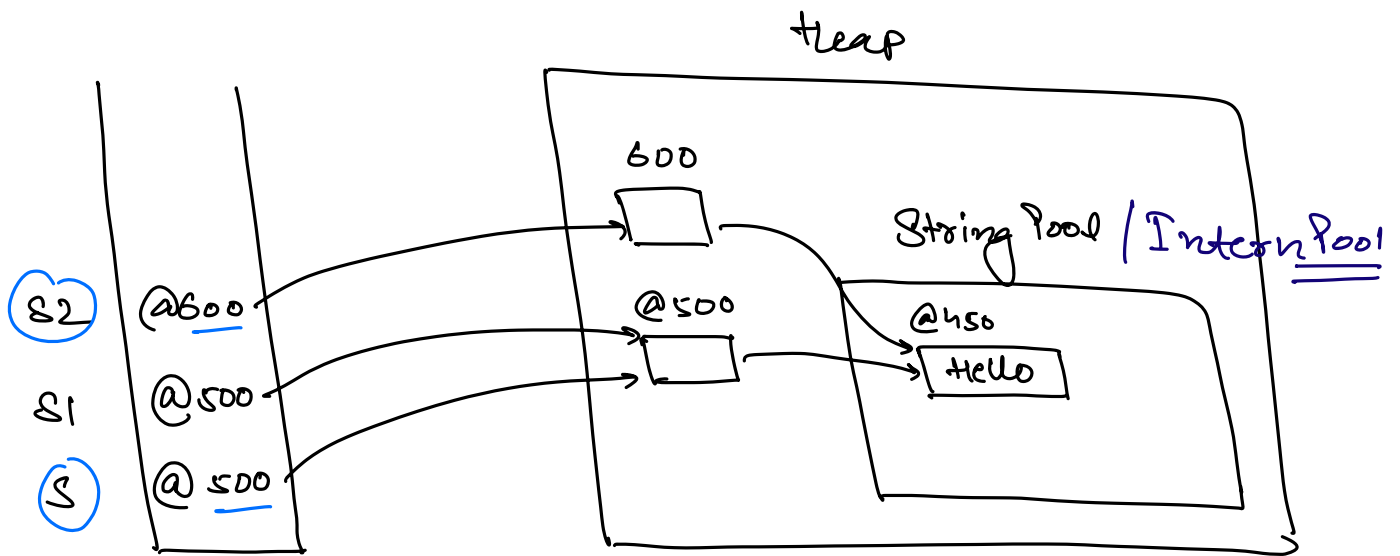
Throw an Exception
if something wrong
has happened.

STRINGS.

↳ Class.

String s = "Hello"

String s1 = "Hello"



⇒ Saving Memory

String s2 = new String("Hello");

⇒ Strings in JAVA are Immutable.

String s = "Hello"

String s2 = new String("Hello");

if (s == s2) {
 ≡
 ≡
 ≡
} ⇒ false.

== : Only compares the references.

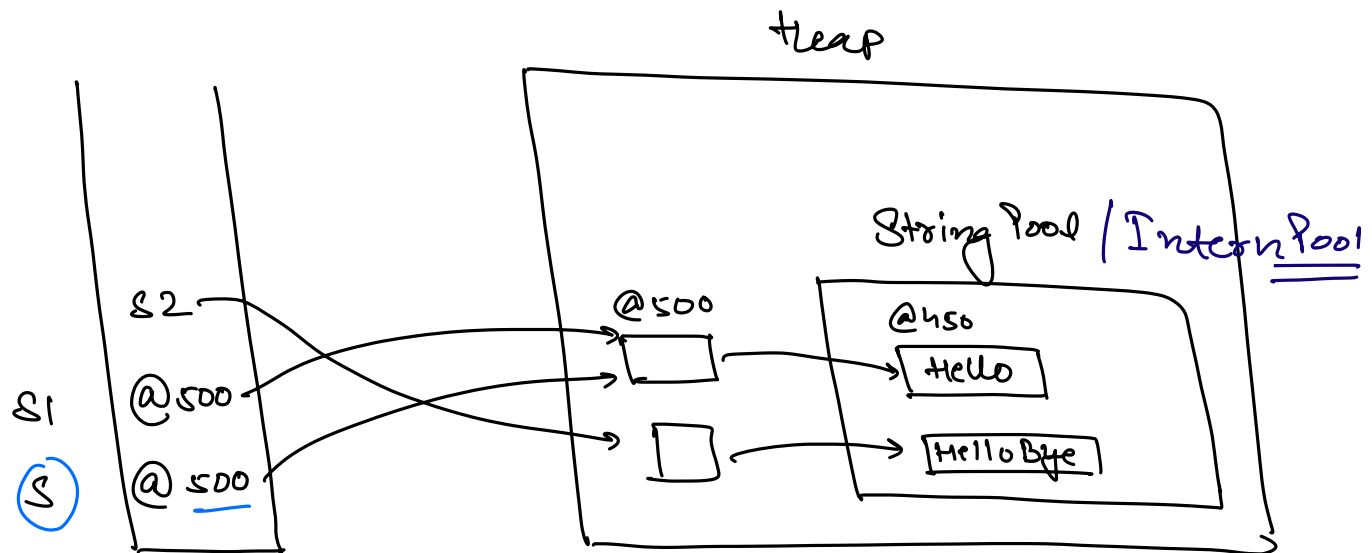
.equals(—)

if (s.equals(s2)) {
 ≡
 ≡
}

if (s == s1) {
 ≡
 ≡
}



⇒ String s = "hello";
String s1 = "hello";
String s2 = s.concat("Bye")

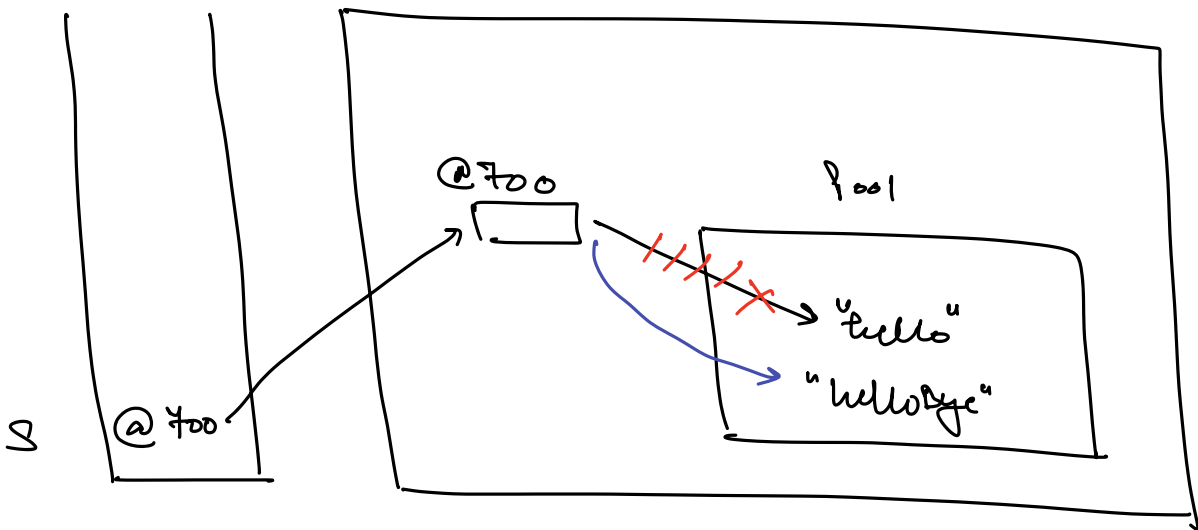


StringBuffer | String Builder.

↓

Array of Character.

String $s = \text{"hello"};$



$s = s.concat(\text{"Bye"});$

Ex:

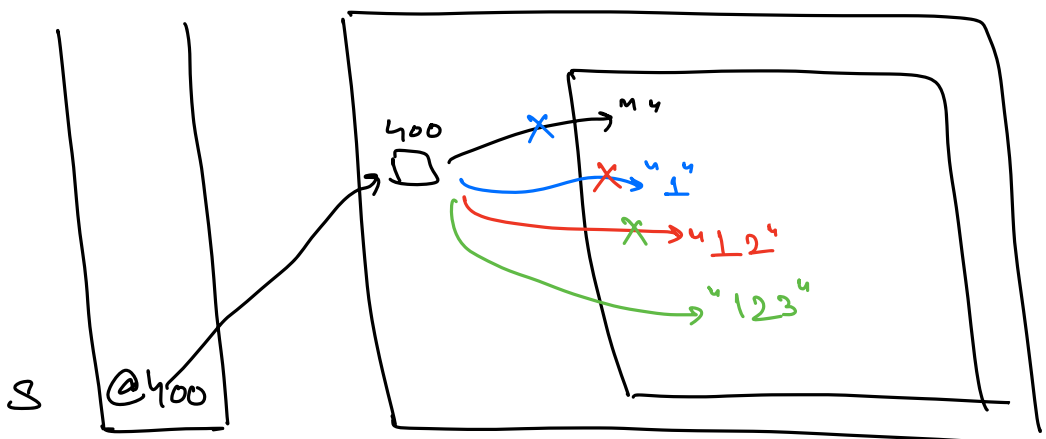
String $s = \text{" "};$

for ($i = 1; i \leq n; i++$) {

$s = s + i$

TC:

3
cout(s)



Iterations : $1 + 2 + 3 + \dots + n$

$$\underline{\underline{O(n^2)}}$$

_____ *